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Evaluating Reservoir Communication in a Naturally Fractured Reservoir: An Integrated Surveillance Case Study Using Tracers, Dynamic Modelling and Flow Diagnostics

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Abstract

This paper presents an integrated study aimed at evaluating reservoir communication in a naturally fractured system. The key objective is to determine the extent to which natural fractures influence reservoir performance and flow connectivity in the initial seven years of production and injection. The study applies a combination of tracer diagnostics, dynamic modelling, and reservoir surveillance to assess whether the system initially behaves in a fracture-dominated or matrix-dominated manner, and to support development decisions to maximize recovery.

A multi-disciplinary workflow was deployed early in the field's development, integrating image logs, tracer diagnostics, dynamic reservoir modelling, pressure transient analysis and injection surveillance. Tracers tracked inter-well communication and breakthrough timing, while Residence Time Distribution model clarified matrix versus fracture flow. Dynamic modelling with streamline simulation, calibrated using early production and pressure data, helped assess reservoir response. Flow diagnostics such as rate analysis and Hall plots characterized flow regimes and injection response. Integrating these datasets enabled cross-verification of reservoir behaviour in a naturally fractured setting, enhancing confidence in connectivity interpretation and performance prediction.

All three techniques consistently indicated that the reservoir is matrix-dominated, despite the presence of conductive and resistive natural fractures. Tracer results showed well-behaved, delayed, and predictable breakthrough, suggesting limited fracture connectivity or channelling. Dynamic reservoir simulation results matched historical performance only when matrix properties were the primary control on flow. Flow diagnostics further confirmed matrix-dominated response, with pressure transient and rate data lacking typical fracture flow signatures such as rapid pressure propagation or bilinear flow. Tracer residence time and breakthrough were found to be controlled by well spacing, injection rate and uptime, with no signs of unexpected or rapid water cut typically associated with natural fractures. Hall plots indicate most injectors encountered resistance to flow, which is contrary to expectations for a fractured reservoir with enhanced permeability. These findings demonstrate that while natural fractures exist and contribute to heterogeneity, they do not adversely affect sweep efficiency, pressure support, or early production performance. The reservoir exhibits stable communication and controllable injection response, supporting field development

through conventional matrix-driven strategies. The consistency of interpretation across tracers, reservoir modelling, and diagnostic methods adds robustness to the conclusion and suggests a nondetrimental influence of natural fractures on initial well placement, spacing, and waterflood design decisions. However, with continued production and injection, influence of natural fractures is expected to play a role as the dynamic conditions change within the reservoir.

This study provides a greenfield example where an integrated diagnostic approach demonstrates initial matrix-dominated behaviour in a fractured carbonate reservoir. It challenges the assumption that fractures inherently complicate reservoir management. Influence of natural fractures can be delayed and better managed by avoiding aggressive operating conditions such as high offtake rates and injection close to fracture pressure. The methodology and insights presented offer a transferable knowledge for similar fields where fracture influence is uncertain, helping operators to make data-driven decisions on reservoir connectivity and development planning.

Introduction

Understanding reservoir communication is essential in naturally fractured systems, where flow behavior is influenced by both matrix and fracture properties. A key uncertainty in early field development is the extent to which natural fractures affect fluid movement, pressure support, and recovery efficiency. Traditionally, fractures are associated with complex flow regimes and potential sweep inefficiencies. However, this study presents a contrasting outcome from a greenfield carbonate reservoir with a marginal pool with seven years of history.

Using an integrated surveillance approach, including tracer diagnostics, dynamic modeling, and injection / production flow analytics— we evaluate the role of fractures in controlling production and injection. The objective is to determine whether the reservoir exhibits fracture-dominated or matrix-dominated flow, and how this impacts decisions for optimized development.

Our results demonstrate that the reservoir, while naturally fractured, behaves in a matrix-dominated manner particularly in the early production and injection phase of the field. The surveillance approach not only reduced early uncertainty but also enabled confident planning of well spacing and waterflood strategies. With continued production and injection, natural fractures are likely to be activated if the offtake and injection conditions remain aggressive.

Methodology and Field Context

Field Setting and Objectives

The field is a greenfield carbonate development in the shallow waters, offshore Abu Dhabi, characterized by low to moderate matrix permeability, good storage capacity, and widespread natural fractures identified via core and image logs. The target reservoir consists of layered and heterogeneous Jurassic carbonates encountered at about 10,000ft subsea. The field is a four-way structural closure with about 26km² area and 340ft structural relief, with an inactive edge aquifer.

After three years of primary recovery, the field has been under a peripheral water flood for the past three years. It is currently in the voidage filling stage; downhole gauges supporting a correction of the pressure decline experienced during the early natural depletion. [Figure 1](#) shows the location of the field.

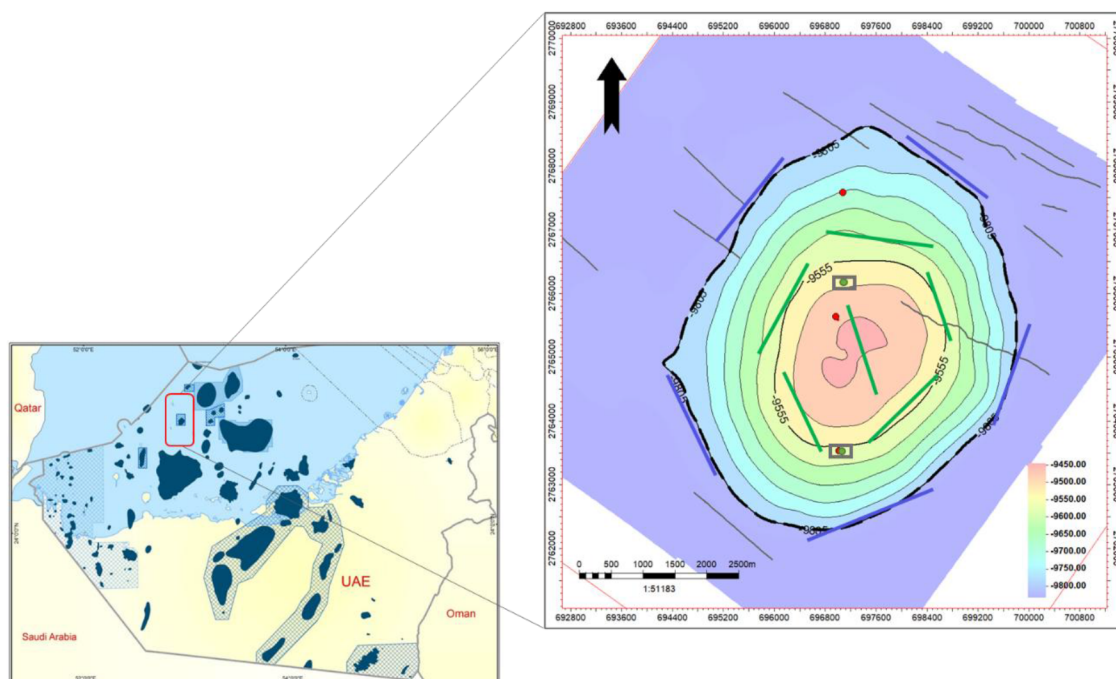


Figure 1—Location map of the candidate field, in ~80ft shallow waters, offshore Abu Dhabi. Three exploration wells are shown in red circles and structural spill in dashed black line. Planned horizontal oil producers and peripheral water injectors are shown in green and blue respectively. Two wellhead towers are shown as grey rectangles.

Reservoir surveillance aimed to (1) Quantify reservoir communication, (2) Assess the flow contribution of matrix versus fractures, (3) monitor injection and production performance and (4) optimize reservoir management. This was achieved by deploying a multidisciplinary diagnostic workflow early in the field life.

Figure 2 offers an overview of surveillance methodologies including tracer diagnostics, dynamic modeling, and flow diagnostics.

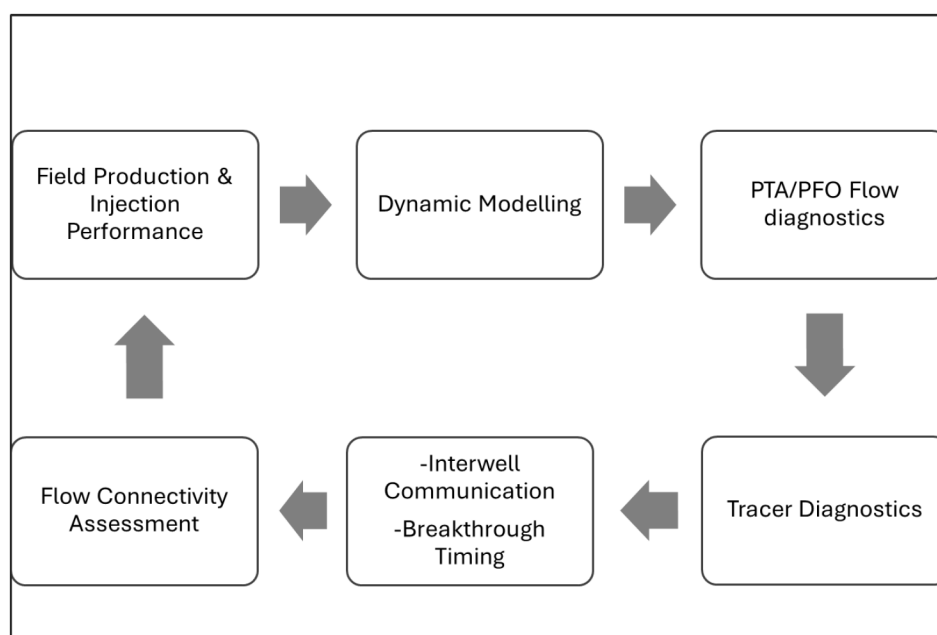


Figure 2—Overview of surveillance methodologies.

Natural Fractures

High angle to near vertical natural fractures are present in the reservoir. Due to the high angle, fractures are less frequent in the cores while they have been adequately intersected in the horizontal wells as seen in the image logs. Figure 3 shows the relative abundance of conductive and resistive fractures across key reservoir layers. Note the fracture density in the heel versus toe area of two wells below.

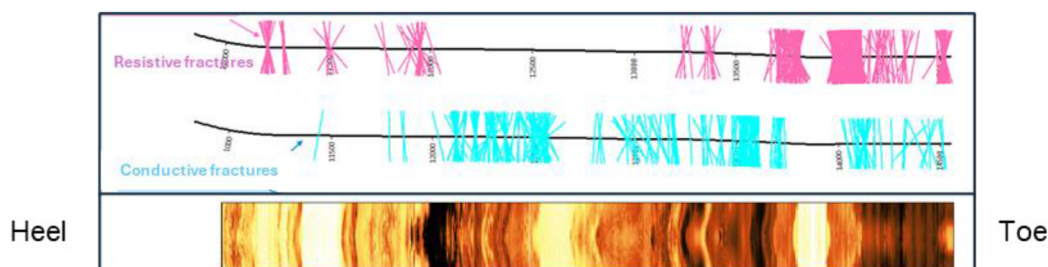


Figure 3—Example of natural fractures encountered along a horizontal well. Compare this with the production performance in Fig.4.

Production Performance

Production performance of the example well is shown in Figure 4. Note the gradual decline and uniform ramp up with no abrupt water cut suggesting matrix dominant flow despite having encountered significant number of natural fractures as shown in Fig.3.

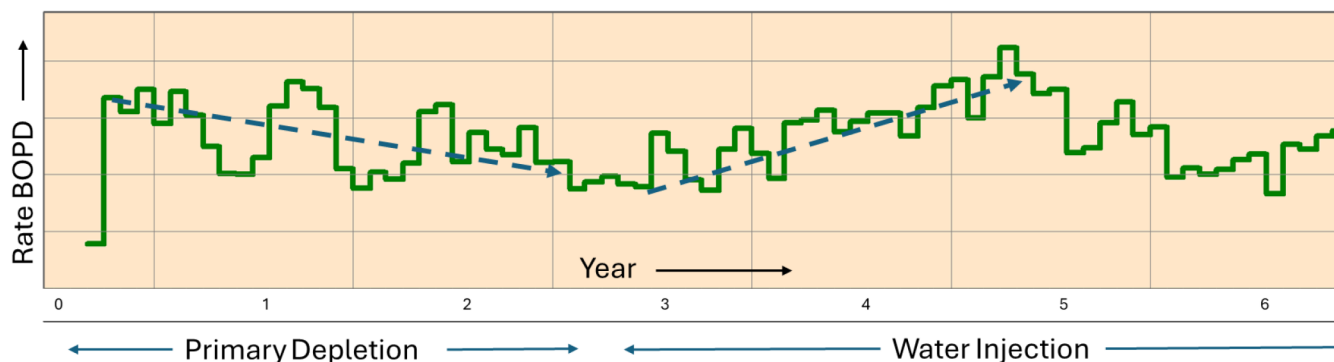


Figure 4—Production performance example well with presence of natural fractures. Note the primary depletion phase for two and half years followed by water flooding. Variable offtake is a function of well uptime and regulatory compliance mandates. Initial production decline is followed by steady production ramp up from the injection support (blue dashed lines). Note no abrupt water breakthrough is seen even after 6 years of production and ~3 years of water flooding suggesting no adverse impact from natural fractures during the early development of this field.

Injection Performance

Hall plot of cumulative water injected versus cumulative injection pressure is shown in Figure 5. It offers relative performance of individual injectors. Clearly several injectors are facing resistance as indicated by their slope above 45°. This is contrary to the expected response of naturally fractured systems that facilitate flow with enhanced permeability.

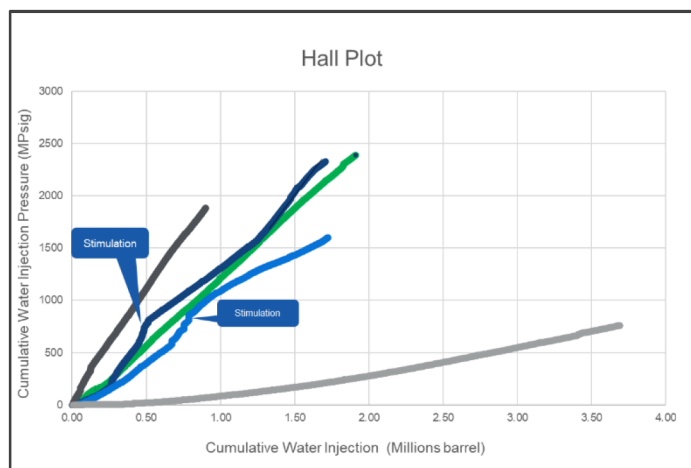


Figure 5—Hall Plot showing relative injection performance of the water injectors. The injector in green indicates stable and normal injection. Higher slope for some others indicates positive skin and resistance to flow. The anomalous low slope in light grey suggests possible channeling or injection out of zone for one injector in the north.

Flow Diagnostics

Complementary production and injection surveillance techniques were applied, including pressure transient analysis (PTA), rate diagnostics, Pressure Fall Off tests (PFOs), Hall Plot, and streamline flow mapping. These diagnostics provided multi-angle evaluation to validate flow pathways and communication behavior which pointed to matrix dominated flow regime. Figure 6 shows an example of a typical PTA response from one of the producers, which shows matrix dominated flow with no sign of fracture contribution.

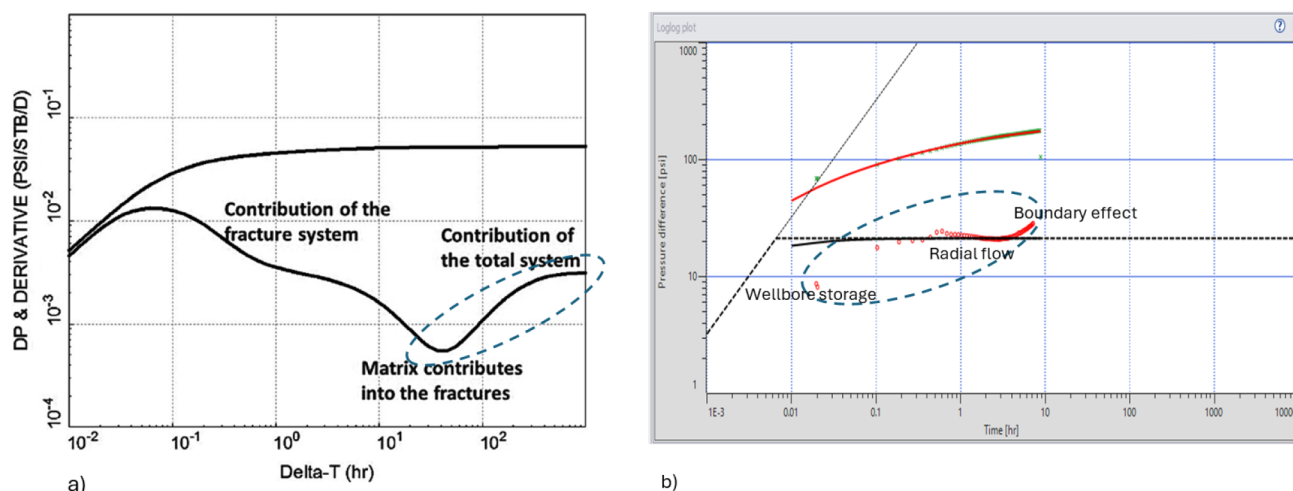


Figure 6—Pressure transient analysis plots a) conceptual and b) actual response of a producer well. Note the derivative plots highlighted in dashed green suggests matrix dominated radial flow with a boundary effect and without any sign of fracture contribution.

Tracer Diagnostics

Single-phase, non-reactive chemical tracers with unique ID were injected into all five water injectors. Sampling was conducted in all the producers at high frequency for six months followed by sampling at monthly intervals. Key measurements included breakthrough timing, peak concentration, and tailing behavior. Tracer response was used to infer the strength and nature of inter-well communication and to identify whether channeling or fracture-dominated transport was present. Figure 7 shows the breakthrough timing of 3 well pairs that range from 75 to 545 days. Holistic analysis reveals the breakthrough time is a function of source and target well spacing, injection rate and well up time.

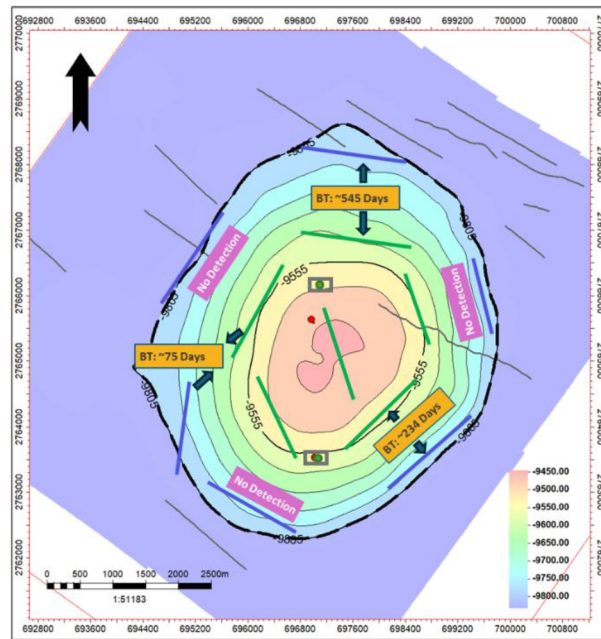


Figure 7—Producer (green) - Injector (blue) well pairs showing tracer breakthrough timing for 3 pairs. Also shown are the remaining injectors whose tracers have not yet been detected by any producer.

The well with fastest tracer breakthrough is discussed below. This well is in proximity, just about 300ft away, from the toe of the source injector, and its tracer return profile shows some sign of fracture response. Note the tracer return conceptual curves compared to actual tracer return profile in this well (Figure 8). The three conceptual tracer return profiles illustrate the following flow regimes:

1. Delayed breakthrough (matrix-dominated response)
 - o Smooth, gradual increase, later peak
 - o Indicates tracer is moving slowly through the matrix, with more dispersion and pore-level diffusion
 - o Typical of low-mobility or tight formations
2. Intermediate response (Mixed matrix-fracture behavior)
 - o Moderate timing, dual-peak or shoulder
 - o May reflect dual porosity behavior where tracer flows initially through fractures and the matrix contributes later. imbibition or exchange
 - o Fractures act as highways for early arrival. Matrix feeds the tracer slowly over time
3. Early breakthrough (Fracture-dominated response)
 - o Sharp rise and fast peak
 - o Suggests direct, fast flow paths (e.g. through fractures or high-perm streaks)
 - o Often observed in fractured or highly heterogeneous reservoirs

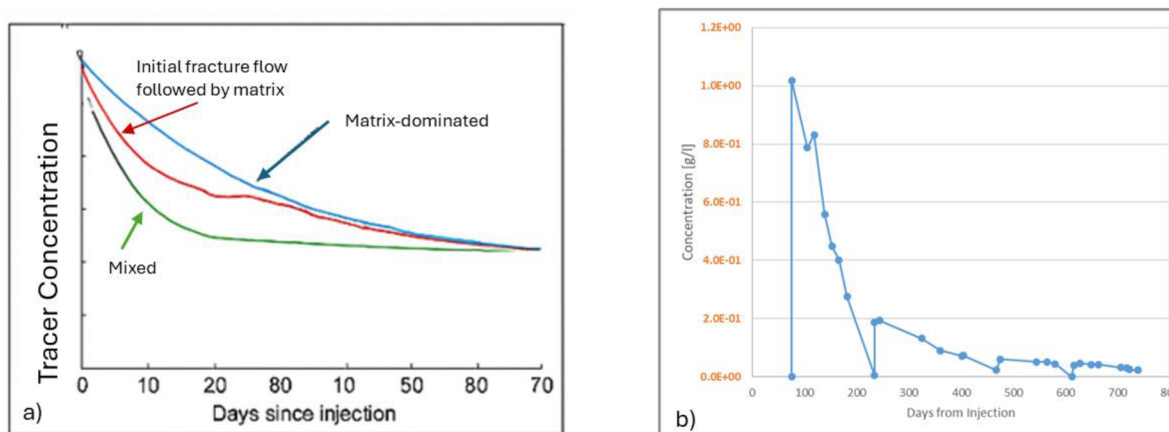


Figure 8—a) Conceptual tracer return profiles illustrating flow regimes compared to b) actual profile seen in a well with tracer breakthrough. Fig. 8a Blue curve - matrix-dominated flow: delayed, smooth return profile with broad dispersion, Red curve - early breakthrough via fractures followed by gradual tracer contribution from the matrix, Green curve - mixed flow regime: a more balanced hybrid flow where neither fractures nor matrix overwhelmingly dominate - a smoother "blended" response. The actual response in 8b corresponds to the red curve in 9a.

The actual tracer return profile exhibits a sharp initial rise suggesting fracture-dominated early flow. Tracer travels quickly through connected fractures or high-perm streaks or due to proximity of the source injector indicating rapid inter-well communication via direct paths. This is followed by the hump and extended tail suggesting matrix contribution. Tracer begins to arrive from slower matrix-dominated regions and continues to dominate the flow. This can be due to imbibition, diffusion into matrix or fracture-to-matrix transfer.

Figure 9 shows the residence time distribution (RTD) model results revealing matrix dominant flow after initial fracture influence. The early breakthrough typically relates to fracture or high perm streak-dominant response. The RTD results complement the actual tracer return profile shown in Fig. 8b but provide further insight into the events and relative quantification of flow regimes. Assessment of sweep efficiency and swept volume suggests about 6% of the injected volume has been produced in this well and the remaining 94% is sweeping the reservoir.

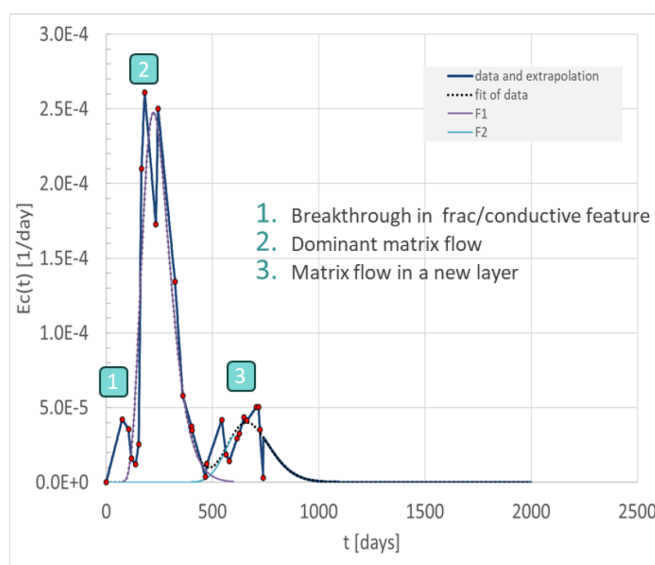


Figure 9—Residence time distribution (RTD) model showing 1) initial tracer breakthrough via fracture/conductive feature followed by 2) matrix dominated flow and 3) a late matrix breakthrough in a new layer.

Dynamic Reservoir Modelling

A matrix-porosity simulation model was built and history matched using early production and pressure data. The model included matrix parameterization from core and log data, and tracer transport to validate inter-well connections.

The objective was to assess the contribution of matrix flow and to calibrate the dynamic model using observed tracer and production data. Figure 10 presents a comparison of tracer breakthroughs in three wells with the onset of water cut predicted by the model. Under matrix-dominated flow conditions, tracer breakthrough closely aligns with the onset of water cut, supporting the model's representation of flow behavior.

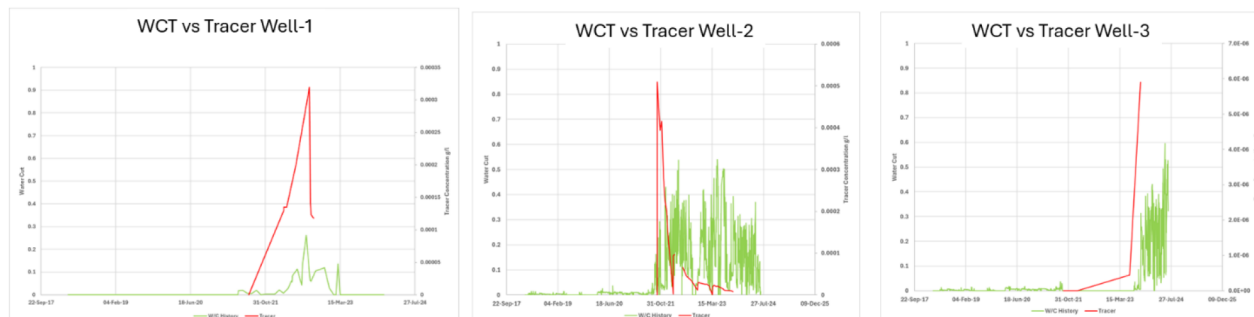


Figure 10—Tracer breakthrough is seen to correspond to the onset of WCT from dynamic model with a few days difference. Tracer concentration peaks correspond to the first WCT peak and then decline due to dilution effect.

Fracture sensitivity was evaluated by adjusting the skin factor (from -7 to -10) to simulate the influence of natural fractures. Figure 11 illustrates a well history match assuming fracture presence. Notably, the fracture-assisted flow deviates from the actual production history, triggering early water breakthrough immediately after water injection begins. This premature breakthrough compromises pressure support, as the well quickly becomes water loaded.

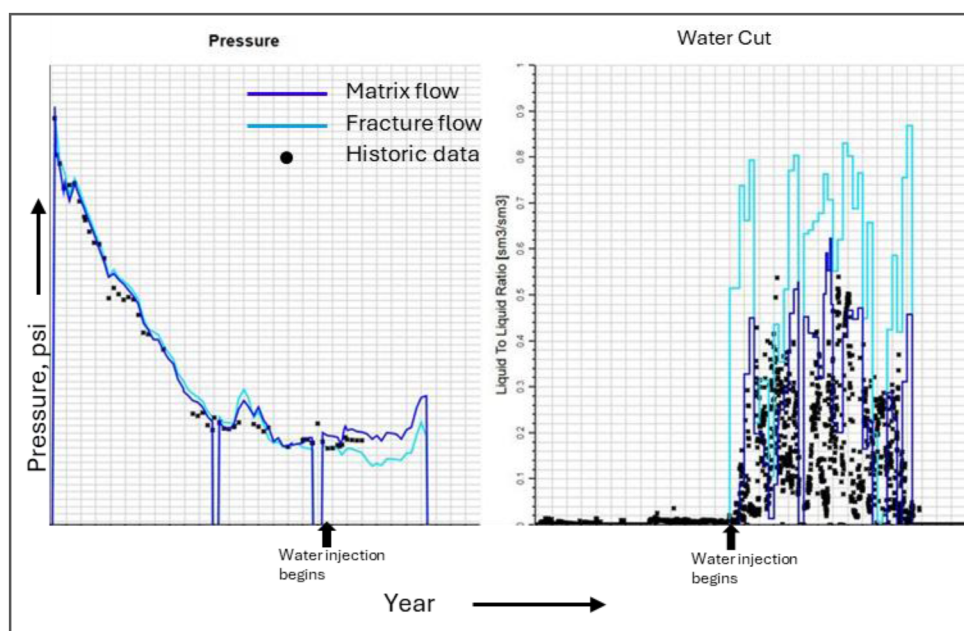


Figure 11—Dynamic model sensitivity of matrix and fracture flow compared to historical data. Note the deviation of fracture flow from historical pressure and water cut as injection begins. Whereas the matrix dominant flow closely matches the pressure and water cut observations. The model is fluid-rate constrained.

Dynamic modelling has been utilized to predict the reservoir performance, monitor flood front and recommend measures to maintain production/injection balance. Current injection volume offers a field VRR of 1.2 to 1.4, considered sufficient to maintain the reservoir pressure above the bubble point. Figure 12 shows the streamlines connecting the oil producers and water injectors. They demonstrate matrix dominated flow regime. In a matrix-dominated system, streamlines move from high to low pressure, their paths are relatively smooth and continuous, shaped by the permeability field and pressure gradients. Whereas in fractured systems, fractures act as high- permeability conduits, so streamlines tend to be channeled through them.

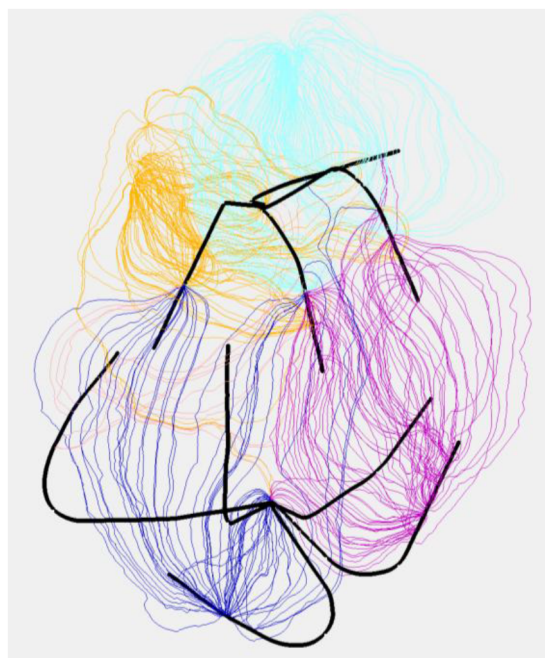


Figure 12—Streamlines showing OP-WI connection in the dynamic reservoir model with matrix dominated system.

Results and Interpretation

Tracer Response Indicates Matrix-Dominated Behaviour

The tracer test results revealed a flow regime predominantly governed by matrix permeability rather than fracture networks. Tracer breakthroughs were observed in a few offset producers; however, the return profiles were notably delayed, symmetric, and exhibited dispersion characteristics except one instance where a rapid breakthrough was observed. These features are indicative of diffusive transport through the porous media, controlled by the intrinsic properties of the rock matrix. The absence of early or asymmetric tracer arrivals—often associated with high-conductivity fractures or preferential flow channels—suggests that the fracture network, while present, did not serve as the primary conduit for fluid flow. In systems dominated by fracture flow, tracer returns typically show rapid breakthrough, sharp peaks, and skewed profiles due to the high velocity and low residence time within fractures. The lack of such signatures in this study implies that the fractures are either poorly connected or possess low transmissivity, acting more as localized heterogeneities than as dominant flow paths. This interpretation is further supported by the spatial distribution of tracer recovery, which was consistent with a matrix-dominated sweep rather than a fracture-controlled anisotropic flow.

Dynamic Modelling Validates Limited Fracture Influence

Numerical reservoir simulation was employed to validate the flow mechanisms inferred from tracer data. History matching and tracer transport modelling were conducted using single-porosity framework and sensitivity of fracture transmissibility through high negative skins.

The reference model prioritized matrix flow which yielded results that closely matched field history. The best-fit scenario incorporated moderate range of matrix permeability (on the order of 10-100 millidarcies), no fracture conductivity, and a dominant matrix pore volume. This configuration successfully reproduced the delayed tracer returns and the gradual production buildup, reinforcing the conclusion that matrix flow is the dominant transport mechanism.

Sensitivity analyses further confirmed that increasing negative skin to simulate fractures led to significant deviations from observed data, underscoring the limited role of fractures in controlling fluid movement. By assigning fracture influence flow as a sensitivity, the model failed to replicate the observed water breakthrough times, pressure and production trends. The model instead predicted premature water arrival, reduced pressure support and underestimated recovery, indicating an unrealistic representation of the subsurface flow dynamics. Flow maps generated from rate, pressure and saturation forecasts over time confirmed that the injected fluids were sweeping the reservoir uniformly, without signs of channeling or short-circuiting. The absence of directional anisotropy in the flow patterns further supports the conclusion that the reservoir is behaving as a matrix-controlled system with minimal fracture influence.

The calibrated model also demonstrated consistent pressure behavior and sweep efficiency across the reservoir, aligning with the diagnostic results from injection and production data. In summary, simulation results showed that history and tracer matches were only achieved when matrix flow properties were dominant.

Flow Diagnostics Support Matrix-Controlled System

Production and Injection well performance and pressure transient analysis (PTA) provided additional evidence of matrix-dominated behavior. Step-rate tests and fall-off analyses revealed stable injectivity indices and pressure responses that lacked the hallmarks of fracture-dominated systems, such as sharp pressure drops, rapid pressure propagation, or bilinear flow regimes. Instead, the pressure build-up curves exhibited radial and pseudo-radial flow characteristics, typical of matrix dominant porous media.

Moreover, pressure transmission between injectors and producers showed moderate and predictable pressure communication, with time-lagged responses indicative of diffusive pressure propagation through the matrix.

Taken together, the tracer analysis, dynamic modeling, and flow diagnostics converge to a coherent interpretation: the reservoir exhibits early matrix-dominated flow behavior, with fractures contributing only marginally to overall fluid transport and heterogeneity.

Looking Ahead - Increasing Influence of Fracture Assisted Flow Anticipated

We believe the early reservoir response in this case study corresponds to type IV reservoir characteristic where fractures do not provide significant additional storage capacity or permeability but simply create anisotropy (Nelson 2001).

However, as production and injection continue over time, evolving reservoir dynamics may lead to the progressive opening of natural fractures, creating preferential flow conduits. If production and injection rates are not carefully balanced and continuously monitored, there is a significant risk of produced water recycling or rapidly displacing oil production assisted by existing natural fractures - a condition that would be difficult to remediate once established.

Dynamic reservoir response in this case would change from type IV to type III as shown in Figure 13. Once individual wells produce mostly water, costly reservoir management interventions may be required

to maintain production. It is important to manage the reservoir with certain proactive measures as below to prevent such a situation, and help extend production and maximize oil recovery:

- Produce under low pressure drawdowns and sustainable rates
- Shut in or recomplete the individual wells with high water production
- Shutting in the entire reservoir to allow fluid contacts to re-equilibrate
- Resort to intermittent water injection
- Spread production over broader lateral area to prevent water channelling by careful well placement or later side tracks, thus minimizing local drawdowns
- Modify injection pattern and well placements

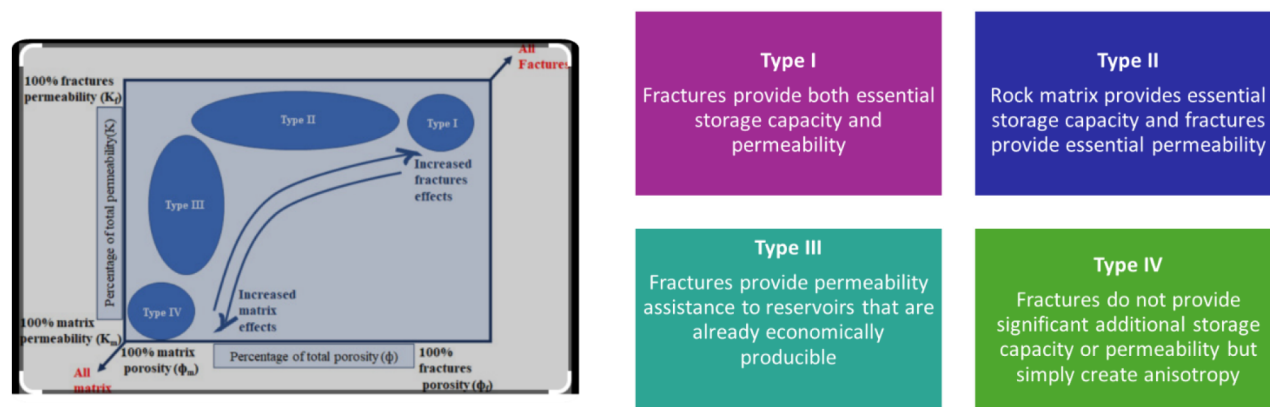


Figure 13—Types of fractured reservoirs and relative impact of matrix and fractures to flow dynamics (after Nelson 2001)

Figure 14 shows the production performance and water cut of a well that experienced water breakthrough after three years of primary depletion followed by two years of voidage fill. The pool is relatively small, and the well is 1.5 km from the nearby injector. Note the rise in water production, which is a combined effect of initial depletion followed by strong injection rate and rapid voidage fill in a relatively small pool with presence of natural fractures. We believe this is an early sign of natural fractures starting to influence the flow dynamics after achieving the voidage fill.

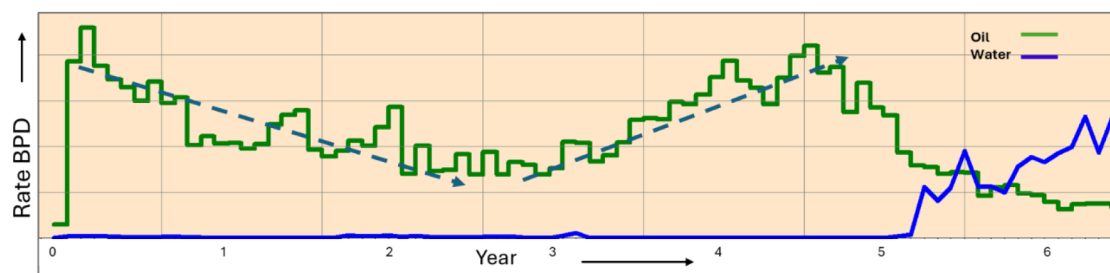


Figure 14—Production performance and water cut response of a well that shows injection water breakthrough after about 2 years of water flooding. Note the relatively rapid water breakthrough after 2 years of high-rate injection suggesting natural fractures may have been activated under changing stress conditions, altering the fluid flow.

Conclusions and Recommendations

This greenfield development study provides fascinating evidence that the presence of natural fractures does not inherently lead to complex or adverse flow behavior early in the field development. Contrary to conventional expectations in fractured reservoirs, the fluid dynamics within this asset are predominantly

governed by matrix permeability. The matrix controls the fluid movement, sweep efficiency, and pressure communication, even in the presence of geologically identifiable fractures. This insight challenges the assumption that fractures always dominate flow regimes and highlights the importance of such validation in early field development. However, fractures will be activated and start to influence fluid flow if the operating conditions remain aggressive, that is, high offtake together with injection close to fracture pressure.

Key Conclusions

- Tracer analysis confirmed matrix-dominated flow behavior in the early production life, with return profiles exhibiting delayed, symmetric, and diffusive characteristics. There were no indications of preferential pathways, channeling, or early breakthrough typically associated with high-conductivity fractures.
- Reservoir simulation required matrix control to achieve accurate history and tracer breakthrough matches. Model sensitivity capturing fracture flow over-predicted water arrival and misrepresented pressure and recovery, underscoring the limited role of fractures impacting initial fluid flow in this system.
- Flow diagnostics revealed stable and predictable reservoir response. Pressure transient analysis and flow mapping showed uniform pressure propagation and sweep patterns, with no evidence of fracture-induced anomalies to date.
- As field production and injection operations continue, there is an increasing likelihood that existing fractures will begin to influence fluid flow dynamics—particularly under sustained aggressive operating conditions. These include elevated injection pressures approaching or exceeding fracture pressure, high voidage replacement ratios, and substantial offtake rates. Such conditions can enhance fracture activation and connectivity, potentially altering flow paths and impacting sweep efficiency and reservoir performance.

Recommendations

- Validate fracture influences empirically through early surveillance. Relying solely on geological or petrophysical indicators may lead to mischaracterization of flow regimes. Incorporating tracer tests, pressure diagnostics, and dynamic modeling early in the field life can provide a more accurate understanding of the reservoir behavior.
- Standardize the integration of tracer analysis, reservoir simulation, and injection diagnostics in greenfield developments. This multidisciplinary approach enhances reservoir characterization and reduces uncertainty in development planning.
- Optimize development strategies based on matrix-dominated assumptions when supported by surveillance data. Well placement, completion design, and injection schemes should be tailored to leverage matrix permeability, while avoiding aggressive water injection which might open the natural fractures.
- To maximize recovery, avoid poorly planned and badly executed injection programs in naturally fractured reservoirs. Instead, implement proactive reservoir management strategies that maintain a balanced approach between pressure support and offtake rates, minimizing the risk of rapid water channeling.

Acknowledgements

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Acronyms

PTA	Pressure Transient Analysis
PFO	Pressure Fall Off test
RTD	Residence time distribution
VRR	Voidage Replacement Ratio
OP-WI	Oil Producer - Water Injector

References

1. Al-Kobaisi, M. S., et al (2012). Integration of Tracer and Streamline Simulation for Interwell Connectivity and Reservoir Management. *SPE Reservoir Evaluation & Engineering*, **15**(5), 536-548.
2. You, L., Geng, L., & Li, H. (2018). *Workflow to Evaluate Waterflood Performance in Fractured Reservoirs Using Tracers and Flow Diagnostics*. SPE ATCE.
3. Thomas, L. K., Corbett, P. W. M., & Jensen, J. L. (1998). Modeling Fractured Reservoirs: The Role of Geology, Geophysics, and Reservoir Engineering. *Journal of Petroleum Technology*, **50**(6), 482-492.
4. Kazemi, H. (1976). Pressure Transient Analysis of Naturally Fractured Reservoirs. *SPE Journal*, **16**(6), 451-462.
5. Samantray, A. et al (2023). *Leveraging Strategic Lessons Learned from a Challenging Marginal Field Water Flood Development*, SPE-215977-MS, ADIPEC
6. Loobari, S and Samantray, A. et al (2024). *Marginal Field Water Flood Management Combining Strategic Planning and Cost- Effective Solutions*, SPE-218971-MS, SPE Water Lifecycle Management Conference and Exhibition, 5-7 March 2024 | Abu Dhabi
7. Lotfy M, et al (2020). *First Co-implementation of Inflow and Inter-well Tracers Offshore Abu Dhabi: Towards Cost Efficient Reservoir Monitoring and Management*, SPE-203132-MS, ADIPEC
8. Nelson, R.A. 2001. *Geologic Analysis of Naturally Fractured Reservoirs*, 2nd edition, Gulf Professional Publishing, Houston.